

Roma Patel

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EDUCATION

- **Brown University** Providence, RI
Doctor of Philosophy, Computer Science; GPA: 4.0 Sep. 2018 – August 2022
- **Rutgers, the State University of New Jersey** Camden, NJ
Bachelor of Science in Computer Science; GPA: 3.95 Sep. 2016 – May. 2018
- **University of Pennsylvania** Philadelphia, PA
Undergrad Research in Computer Science; 4 Course Units Sep. 2017 – May. 2018
- **Mumbai University** Mumbai, India
Bachelor of Science in Computer Science Sep. 2014 – May. 2016

WORK EXPERIENCE

- **Google DeepMind (Senior Research Scientist)** San Francisco, CA
Research Scientist September 2025-current
 - **Research:** Natural language processing, safety, alignment, interpretability, reinforcement learning methods
- **Google DeepMind (Research Scientist)** London, UK
Research Scientist September 2022-2025
 - **Research:** Natural language processing, multi-agent reinforcement learning, reinforcement learning
- **Microsoft Research (Internship)** Seattle, WA (remote)
Turing Academic Program; Led by Dean Carignan and Roy Zimmerman Summer 2021
 - **Projects:** Methods for interpretability and probing of bias in generative language models; and building systems that can detect and control harmful outputs.
- **DeepMind (Internship)** London, UK (remote)
Research Intern - Multi-agent Team; Advised by Dr. Yoram Bachrach Summer 2020
 - **Projects:** Building natural language communication channels for multi-agent systems; and game-theoretic vocabulary-selection algorithms for natural language tasks.
- **Google AI (Internship)** Mountain View, CA
Research Intern - Grounded Language Team; Advised by Dr. Jason Baldridge and Alexander Ku Summer 2019
 - **Projects:** Building multilingual frameworks for instruction-following reinforcement learning agents; and probing agent behaviour for more interpretable reinforcement learning navigation agents.
- **Johns Hopkins University (JSALT Workshop)** Baltimore, MD
Research Intern - Natural Language Processing; Advised by Dr. Ellie Pavlick and Dr. Sam Bowman Summer 2018
 - **Projects:** Building and evaluating language models for general purpose sentence representation; visually grounded learning objectives; probing language models.
- **University of Pennsylvania (Research Assistant)** Philadelphia, PA
Research Assistant - Natural Language Processing; Advised by Dr. Ani Nenkova and Dr. Byron Wallace May 2017 - 2018
 - **Projects:** Structured prediction models for information extraction; neural models to determine ambiguity in natural language text.
- **Princeton University (Research Assistant)** Princeton, NJ
Student and Research Assistant (PACT); Advised by Dr. Rajiv Gandhi Summer 2016
 - **Projects:** Approximation algorithms for NP Hard problems; mentoring students on algorithms problem sets.

LECTURES AND INVITED TALKS

- **Safety and Alignment of Language Models:** M2L Summer School 2025
- **Brown University, CSCI 1460:** Grounded Language Learning
- **Montreal Institute for Learning Algorithms, Seminar:** Grounded Language Learning
- **DIMACS, Centre for Discrete Mathematics and Theoretical Computer Science:** Language Models and Multi-agent Learning
- **Columbia University, Data Science Institute:** Learning from Patterns for Information Extraction for Medical Literature
- **Princeton University, PACT Summer Program:** Network Flows

ACHIEVEMENTS

- **Best Paper Award:** StarSEM 2019.
- **Presidential Fellowship:** Brown University, 2018-2019.
- **Dean's Undergraduate Research Award:** Rutgers University, 2018.
- **Computer Science Achievement Award:** Rutgers University, 2018.
- **Academic Excellence Scholarship:** Rutgers University, 2016.
- **Regional Math Olympiad Finalist:** Placed second in a series of competitive national-level olympiad rounds, 2015.
- **Dean's List:** Dean's List every semester; Rutgers University.
- **Scholarship for Higher Education:** Innovation in Science Pursuit for Inspired Research (INSPIRE)

PUBLICATIONS

- **Syntactic Patterns Improve Information Extraction for Medical Literature:**
Roma Patel, Yinfei Yang, Iain Marshall, Ani Nenkova and Byron Wallace. NAACL 2018.
- **A Corpus with Multi-Level Annotations of Patients, Interventions and Outcomes to Support Language Processing for Medical Literature:**
Benjamin Nye, Jessie Li, Roma Patel, Yinfei Yang, Iain Marshall, Ani Nenkova and Byron Wallace. ACL 2018.
- **Learning Visually Grounded Representations with Sketches:**
Roma Patel, Stephen Bach and Ellie Pavlick. The How2 Challenge: New Tasks for Vision Language. ICML 2019.
- **Looking for ELMo's friends: Sentence-Level Pretraining Beyond Language Modeling:**
Samuel R. Bowman, Ellie Pavlick, Edouard Grave, Benjamin Van Durme, Alex Wang, Jan Hula, Patrick Xia, Raghavendra Pappagari, R. Thomas McCoy, Roma Patel, Najoung Kim, Ian Tenney, Yinghui Huang, Katherin Yu, Shuning Jin, Berlin Chen. ACL 2019.
- **Probing What Different NLP Tasks Teach Machines about Function Word Comprehension:**
Najoung Kim, Roma Patel, Adam Poliak, Alex Wang, Patrick Xia, R. Thomas McCoy, Ian Tenney, Alexis Ross, Tal Linzen, Benjamin Van Durme, Samuel R. Bowman, Ellie Pavlick. StarSEM 2019. **Best paper award!**
- **Planning with State Abstractions for Temporal Non-Markovian Task Specifications:**
Yoonseon Oh, Roma Patel, Thao Nguyen, Baichuan Huang, Ellie Pavlick and Stefanie Tellex. RSS 2019.
- **Learning to Ground Language to Temporal Logical Form:**
Roma Patel, Stefanie Tellex and Ellie Pavlick. SPLU-RoboNLP. NAACL 2019.
- **Robot Object Retrieval with Contextual Natural Language Queries:**
Thao Nguyen, Nakul Gopalan, Roma Patel, Matthew Corsaro, Ellie Pavlick, Stefanie Tellex. RSS 2020.
- **Grounding Language to Non-Markovian Tasks with No Supervision of Task Specifications:**
Roma Patel, Ellie Pavlick, Stefanie Tellex. RSS 2020.
- **On the Relationship Between Structure in Natural Language and Models of Sequential Decision Processes:**
Roma Patel*, Rafael Rodriguez-Sanchez*, George Konidaris. LaREL. ICML 2020.
- **Room-Across-Room: Multilingual Vision-and-Language Navigation with Dense Spatiotemporal Grounding:**
Alexander Ku, Peter Anderson, Roma Patel, Eugene Ie and Jason Baldridge. EMNLP 2020.
- **Game-theoretic Vocabulary Selection for Text Classification Tasks:**
Roma Patel, Ian Gemp, Chris Dyer, Marta Garnelo, Yoram Bachrach. NAACL 2021.
- **Was it "stated" or was it "claimed"? How linguistic bias affects generative language models:**
Roma Patel and Ellie Pavlick. EMNLP 2021.

- **Mapping Language Models to Grounded Conceptual Spaces:**
Roma Patel and Ellie Pavlick. ICLR 2022.
- **Generalising to New Domains by Mapping Natural Language to Lifted LTL.:**
Eric Hsiung, Hiloni Mehta, Junchi Chu, Xinyu Liu, Roma Patel, Stefanie Tellex, George Konidaris. ICRA 2022.
- **Affordance-based robot object retrieval:**
Thao Nguyen, Nakul Gopalan, Roma Patel, Matt Corsaro, Ellie Pavlick, Stefanie Tellex. Autonomous Robots 2022.
- **Hierarchical planning with state abstractions for temporal task specifications:**
Yoonseon Oh, Roma Patel, Thao Nguyen, Baichuan Huang, Matthew Berg, Ellie Pavlick, Stefanie Tellex. Autonomous Robots 2022.
- **Pragmatics in grounded language learning: Phenomena, tasks, and modeling approaches:**
Daniel Fried, Nicholas Tomlin, Jennifer Hu, Roma Patel, Aida Nematzadeh. EMNLP Findings 2023.
- **Skill Generalization with Verbs:**
Rachel Ma, Lyndon Lam, Benjamin A Spiegel, Aditya Ganeshan, Roma Patel, Ben Abbatematteo, David Paulius, Stefanie Tellex, George Konidaris. ICRA 2023.
- **States as Strings as Strategies: Steering Language Models with Game-Theoretic Solvers:**
Ian Gemp, Roma Patel, Yoram Bachrach, Marc Lanctot, Vibhavari Dasagi, Luke Marris, Georgios Piliouras, Siqi Liu, Karl Tuyls. AAMAS 2024.
- **Strategic Foundation Models:**
Denizalp Goktas, Amy Greenwald, Takayuki Osogami, Roma Patel [..] HAL Open Science 2024
- **Whose view of safety? A deep DIVE dataset for pluralistic alignment of text-to-image models:**
Charvi Rastogi, Tian Huey Teh, Pushkar Mishra, Roma Patel, Zoe Ashwood, Aida Mostafazadeh Davani, Mark Diaz, Michela Paganini, Alicia Parrish, Ding Wang, Vinodkumar Prabhakaran, Lora Aroyo, Verena Rieser. NeuRIPs 2025.
- **Value Profiles for Encoding Human Variation:**
Taylor Sorensen, Pushkar Mishra, Roma Patel, Michael Henry Tessler, Michiel Bakker, Georgina Evans, Iason Gabriel, Noah Goodman, Verena Rieser. EMNLP 2025.
- **Nuanced Safety for Generative AI: How Demographics Shape Responsiveness to Severity:**
Pushkar Mishra, Charvi Rastogi, Stephen R. Pfohl, Alicia Parrish, Roma Patel, Mark Diaz, Ding Wang, Michela Paganini, Vinodkumar Prabhakaran, Lora Aroyo, Verena Rieser. TMLR 2025.
- **Evaluating safety in GenAI Image safety annotation tasks from diverse annotators' perspectives:**
Ding Wang, Mark Díaz, Charvi Rastogi, Aida Davani, Vinodkumar Prabhakaran, Pushkar Mishra, Roma Patel, Alicia Parrish, Zoe Ashwood, Michela Paganini, Tian Huey Teh, Verena Rieser, Lora Aroyo. AIES 2025.
- **Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities:**
G Comanici, E Bieber, M Schaeckermann, I Pasupat, Roma Patel [..]. arXiv 2025.
- **Is This Just Fantasy? Language Model Representations Reflect Human Judgments of Event Plausibility:**
Michael Lepori, Jennifer Hu, Ishita Dasgupta, Roma Patel, Thomas Serre, Ellie Pavlick. ICLR 2026.
- **Utility-inspired Reward Transformations Improve Reinforcement Learning Training of Language Models:**
Roberto-Rafael Maura-Rivero, Chirag Nagpal, Roma Patel, Francesco Visin. AAMAS Workshop 2026.
- **Training a population of adversarial neural networks to improve a base neural network:**
Roma Patel, John Agapiou, Marta Abellanas, Andrea Tacchetti. US Patent. 2026. Blog
- **Vocabulary selection for text processing tasks using power indices:**
Ian Gemp, Yoram Bachrach, Chris Dyer, Roma Patel. US Patent. 2026.
- **Empowering inclusive e-deliberation through position trees and dialectical trees:**
Gabriela Díaz, Carlos Chesñevar, Roma Patel, Ana Maguitman. JAIIO 2026.
- **Positive Alignment: Artificial Intelligence for Human Flourishing:**
Ruben Laukkonen, Seb Krier, Chloé Bakalar, Shamil Chandaria, Morten Kringelbach, Adam Elwood, Daniel Ford, Fernando Rosas, Maty Bohacek, Matija Franklin, Nenad Tomašev, Stephanie Chan, Verena Rieser, Roma Patel, Michael Levin, Arun Rao. arXiv. 2026.
- **Detecting and Controlling Sycophancy with Cascading Linear Features:**
Maty Bohacek, Rishub Jain, Nicholas Dufour, Thomas Leung, Chris Bregler, Roma Patel. arXiv. 2026.
- **Pluralis v0.1: Towards a Multicultural, Multimodal, Multilingual Benchmark for AI Risk and Reliability:**
Alicia Parrish, Rajat Shinde, Roma Patel [..]. arXiv. 2026.
- **Long-term Measurements: Towards a Longitudinal Understanding of Human-AI Interactions:**
Nicole Mitchell, Dhruv Agarwal, Maty Bohacek, Remi Denton, Roma Patel. arXiv. 2026.